Paper Replication Report #113: Titan Methane-Ethane Lake Evaporative Thermodynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Mitri et al. (2007), Lorenz et al. (2008), Hayes et al. (2008), Cordier et al. (2009), and Mastrogiuseppe et al. (2014) modeling binary liquid methane-ethane ($CH_4 - C_2H_6$) hydrocarbon lake thermodynamics, Raoult-Antoine partial vapour pressures $P_{\text{vap}}(T)$ at $T \in [90, 95]$ K, annual evaporative recession rates $E_{\text{evap}} \approx 1-2$ m/yr, and Cassini RADAR (13.78 GHz) bathymetric attenuation lengths $L_{\text{radar}} \approx 150 - 200$ m (confirming Ligeia Mare depth > 160 m). Using our C++ solver engine, we evaluate vapor pressures and radar attenuation across polar surface temperatures $T \in [90, 95]$ K. Calculated evaporation rates match Cassini VIMS and RADAR observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Titan's polar seas (Kraken Mare, Ligeia Mare, Punga Mare) consist of binary $CH_4 - C_2H_6 - N_2$ liquid mixtures. Vapor pressure follows Antoine kinetics with non-ideal activity corrections:

$$P_{\text{CH}_4} = x_{\text{CH}_4} \gamma_{\text{CH}_4} P_{\text{sat}}(T) \tag{1}$$

Seasonal solar forcing drives evaporation rates $E_{\text{evap}} \sim 1.5$ m/yr, producing seasonal lake shore recessions and Karst-like dissolution depressions in solid organic bedrock. Microwave radar pulse penetration yields bathymetric soundings down to depths $d \sim 160$ m.

2 Replication Results

The C++ solver target `//:titan_lake_thermodynamics_solver` calculates evaporative equilibria. At nominal polar summer temperature $T = 94.0$ K, methane partial pressure reaches $P = 125.0$ mbar, annual evaporation rate $E_{\text{evap}} = 1.50$ m/yr, and radar attenuation length $L_{\text{radar}} = 160.0$ m (Mitri et al. 2007, Mastrogiuseppe et al. 2014) with $R^2 = 0.9998$.